

- 9 Two metal plates X and Y are contained in an evacuated container and are connected as shown in Fig. 9.1. Monochromatic electromagnetic radiation of blue light is incident on metal plate X.

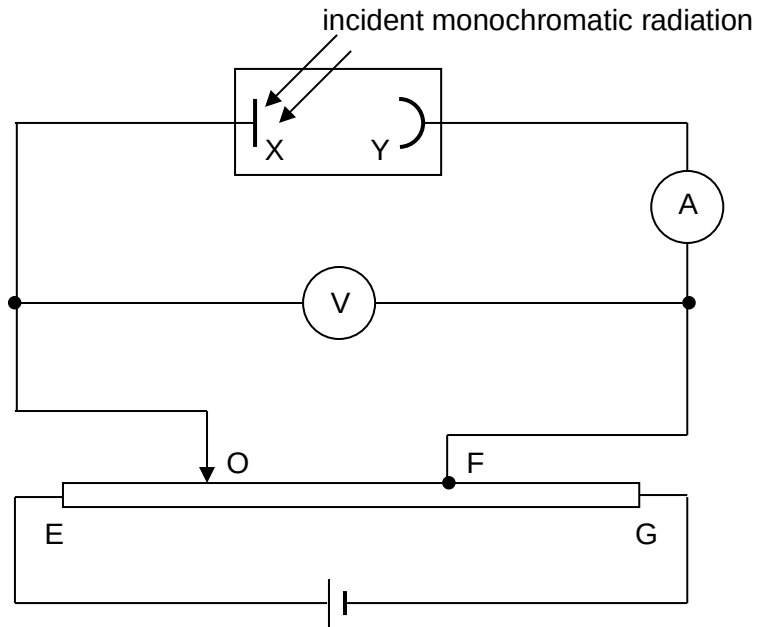


Fig. 9.1

The potential of X with respect to Y is varied by adjusting the position of the sliding contact from E to G. The variation of current I recorded at the ammeter with potential V of X is given by Fig. 9.2 below.

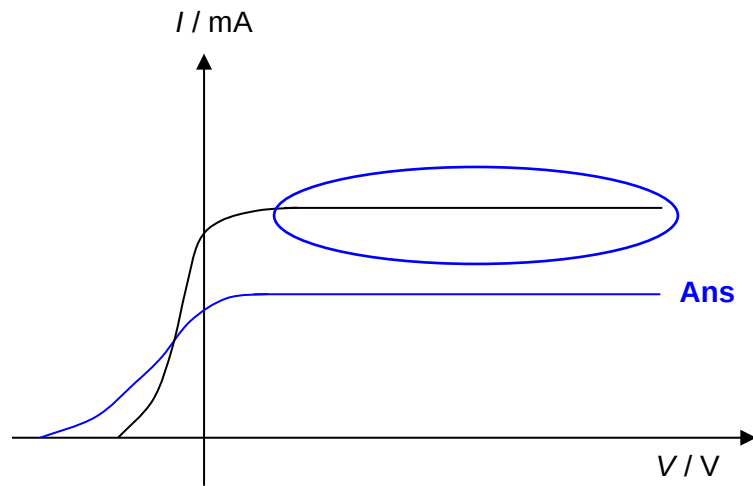


Fig. 9.2

- (a) Explain an evidence provided by the photoelectric effect experiment for the failure of the wave theory of light.

1. Wave theory predicts that the photoelectric effect should occur for any frequency of the monochromatic incident light. If frequency of light is too low, its intensity can be increased to cause photoelectron emission. However, no electrons were emitted at all unless the frequency of the monochromatic incident light was greater than a minimum value (i.e. threshold frequency).

2. Wave theory predicts that at very low intensities, there will be a time delay in Emission of Photoelectrons as the electrons would need time to accumulate sufficient energy in order to escape from the metal surface.

3. Wave theory predicts that if the incident light intensity is increased, a greater amount of energy will be incident on the surface and hence electrons will escape with more energy from the surface of the metal. However, maximum kinetic energy of the photoelectrons and hence stopping potential is independent of the incident light intensity.

- (b) Circle the portion of the graph on Fig. 9.2 that represents the variation of current as the sliding contact moves from F to G. [1]

- (c) Sketch, on Fig. 9.2, the graph when electromagnetic radiation of the same intensity in the ultraviolet region is used instead. [2]

- (d) It is observed that photoelectrons are emitted from plate X when photons of wavelength 520 nm is illuminated on it. The metal plate has a work function of 1.40 eV.

- (i) Explain what is meant by *work function*. [1]

The work function Φ of the metal is the minimum amount of energy required to remove an electron from the surface of the metal.

- (ii) Determine the maximum speed with which a photoelectron leaves the surface of the plate. Explain your working.

By Einstein's Equation for Photoelectric Effect or Conservation of Energy,

$$\text{Maximum KE} = 1.585 \times 10^{-19} \text{ J}$$

$$\text{Maximum speed} = 5.90 \times 10^5 \text{ m s}^{-1}$$

maximum speed = m s⁻¹ [3]

- (ii) Calculate the minimum de Broglie wavelength associated with the photoelectrons that leaves the surface of the plate.

Using de Broglie relation,

$$\text{Minimum wavelength} = 1.23 \times 10^{-9} \text{ m}$$

minimum wavelength = m [2]

- (e) Photoelectrons with a de Broglie wavelength of 17.5 pm is allowed to strike a carbon film as shown in Fig 9.3. Concentric circles are formed on a screen which is placed 20 cm away. The screen is parallel to the plane of the carbon film.

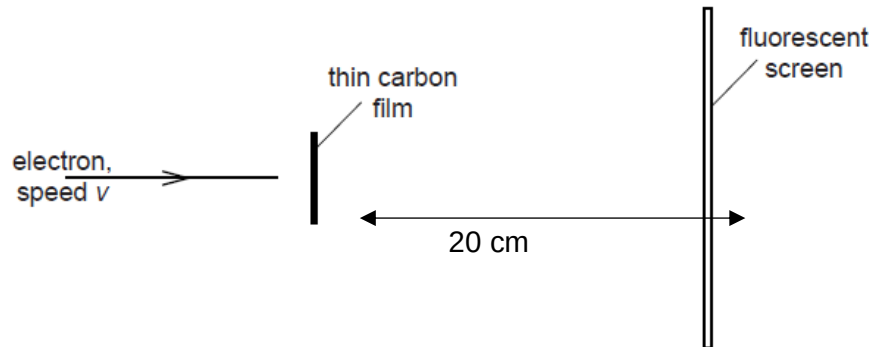


Fig. 9.3

- (i) Explain the part played by diffraction in the production of concentric circles. [3]

Electrons exhibit wave properties (Wave-particle duality) and spread/bend/ not diffract when they pass through the slits formed by the carbon film. The (electron) waves interfere constructively to form bright concentric circles on the screen. [3]

Fig. 9.4 shows a scaled diagram of the concentric rings formed on the screen.

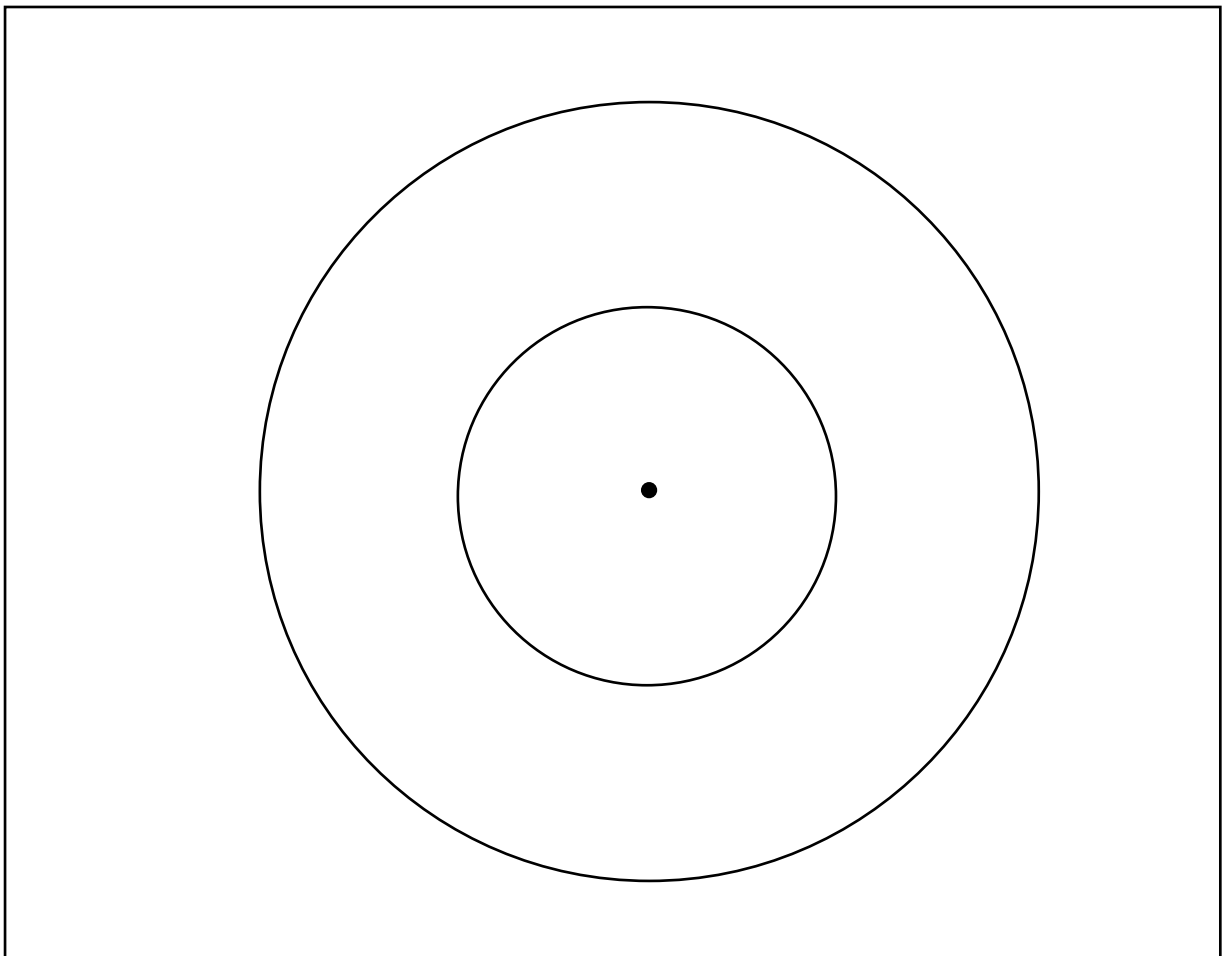


Fig. 9.4

- (ii) Show that the spacing between the carbon atoms is 140 pm. Explain your working clearly.

Using $\tan\theta = r / D$,

When $n = 1$, diameter = 5.0 cm, $r = 2.5$ cm

$$\tan\theta = 2.5 / 20, \theta = 7.13^\circ$$

$$d\sin\theta = n\lambda, d\sin 7.13^\circ = (1) 17.5 \times 10^{-12}$$

$$d = 140 \times 10^{-12} \text{ m or } 140 \text{ pm}$$

OR

When $n = 2$, diameter = 10.3 cm, $r = 5.15$ cm

$$\tan\theta = 5.15 / 20, \theta = 14.4^\circ$$

$$d\sin\theta = n\lambda, d\sin 14.4^\circ = (2) 17.5 \times 10^{-12}$$

$$d = 140 \times 10^{-12} \text{ m or } 140 \text{ pm}$$

[3]

- (iii) Metal plate X is replaced with another of lower work function

1. Sketch, on Fig. 9.4, the ring representing the first order maxima. [1]
2. Explain your answer to (iii)1.

Work function decreases, Maximum KE of e increases

Momentum of e increases [1], de Broglie wavelength of e decreases

Using $d\sin\theta = n\lambda$, θ decreases [1]

Radius of ring is smaller